

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2**

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA17

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### **COURSE OUTCOME COVERED**

**CO1:** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

*Assessment Tool Weightage: 50%*

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**QUESTIONS:** 3

**Duration:** 1 Hour

**Total Marks:**50

### **Annexure A**

### **Constraint-Based Problem Solving**

#### ***Code of Conduct:***

*I **HEMASRINITHA P. (192572179)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

**1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:**

**Constraints:**

**i. D1 cannot work Night shift**

**ii. D2 must work before D3 (Morning &lt; Afternoon &lt; Night)**

**iii. Only one doctor per shift**

**iv. D3 cannot work Morning**

**v. All doctors must be assigned exactly one shift**

**Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule**

**Constraints**

1. D1 cannot work the Night shift.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor can be assigned to each shift.
4. D3 cannot work the Morning shift.
5. Every doctor must be assigned exactly one shift.

**Backtracking Steps**

| Step | Assignment     | Constraint Check                         | Result |
|------|----------------|------------------------------------------|--------|
| 1    | D1 → Morning   | D1 is not on Night                       | Valid  |
| 2    | D2 → Afternoon | D2 is before D3                          | Valid  |
| 3    | D3 → Night     | D3 is not on Morning and D2 is before D3 | Valid  |

Since all constraints are satisfied, no backtracking is required.

**Constraint Verification**

- D1 is not assigned the Night shift.
- D2 (Afternoon) is before D3 (Night).
- Each shift has only one doctor.
- D3 is not assigned the Morning shift.
- All doctors are assigned exactly one shift.

### Final Valid Schedule

| Doctor | Shift     |
|--------|-----------|
| D1     | Morning   |
| D2     | Afternoon |
| D3     | Night     |

### Conclusion:

D1 → Morning, D2 → Afternoon, D3 → Night is the valid assignment obtained using Backtracking Search.

**2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.**

### Constraints:

- i. Movement allowed: Up, Down, Left, Right**
- ii. Cost of each move = 1**
- iii. Cannot pass through obstacles**
- iv. Use Manhattan Distance heuristic**

**Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.**

### Algorithm

1. Start from node **S**.
2. Add **S** to the queue.
3. Remove the first node from the queue.
4. Expand all valid neighboring nodes (Up, Down, Left, Right).
5. Ignore obstacles and already visited nodes.
6. Continue until the Goal node **G** is reached.
7. Trace back the path from **G** to **S**.

### Node Expansion and Queue Updates

| Step | Expanded Node | Queue After Expansion |
|------|---------------|-----------------------|
| 1    | S             | A, B                  |
| 2    | A             | B, C                  |
| 3    | B             | C, D                  |
| 4    | C             | D, E                  |
| 5    | D             | E, F                  |
| 6    | E             | F, G                  |
| 7    | G             | Goal Reached          |

### Optimal Path

$S \rightarrow A \rightarrow C \rightarrow E \rightarrow G$

### Path Cost

- Cost of each move = 1
- Total number of moves = 4

Total Cost = 4

### Manhattan Distance Heuristic

$$h(n) = |x_1 - x_2| + |y_1 - y_2|$$

The Manhattan Distance estimates the distance from the current node to the goal by adding the horizontal and vertical distances.

### Result

- Optimal Path:  $S \rightarrow A \rightarrow C \rightarrow E \rightarrow G$
- Total Cost: 4

**3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.**

### Constraints:

- ☐ The robot can move up, down, left, right

- ☐ Some cells are blocked (obstacles) and cannot be traversed
- ☐ The robot has limited sensing capability (can only observe adjacent cells)
- ☐ Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- ☐ The robot must minimize total cost while ensuring safe navigation
- ☐ The robot must update its knowledge dynamically (online search scenario)

**Tasks:**

**a) Analyze all the given constraints and explain how they affect the robot's decision-making process.**

**b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.**

**c) Demonstrate how the robot applies AI concepts such as:**

- ☐ Intelligent agents
- ☐ Rationality
- ☐ Environment type

**d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.**

**a) Analysis of the Constraints**

- The robot can move only **up, down, left, and right**.
- It cannot move through **blocked cells (obstacles)**, so it has to find another route.
- The robot can see only the **adjacent cells**, so it does not know the entire building at the beginning.
- Every move costs **1**, and entering a **risky zone** adds an extra cost of **2**. Therefore, the robot should choose the safest and least expensive path.
- As the robot explores, it learns more about the environment and updates its decisions based on the new information.

**b) Solution Approach (Online Search using Uniform Cost Search)**

1. The robot starts from the **Start (S)** position.
2. It checks the neighboring cells and identifies the possible paths.
3. It ignores blocked cells and calculates the total cost for each available path.

4. It always chooses the path with the **lowest cost**.
5. Whenever it discovers new obstacles or risky areas, it updates its map and changes its route if necessary.
6. This process continues until the robot safely reaches the **Goal (G)**.

### c) AI Concepts Used

**Intelligent Agent:** The rescue robot acts as an intelligent agent because it senses its surroundings, makes decisions, and moves towards the survivor without human intervention.

**Rationality:**

The robot behaves rationally by selecting the safest path with the minimum total cost while avoiding unnecessary risks.

**Environment Type:**

- **Partially Observable:** The robot can only see nearby cells.
- **Dynamic:** The robot discovers new information while moving.
- **Sequential:** Each move affects the next decision.
- **Single Agent:** Only one rescue robot performs the task.

### d) Justification

An Online Search Agent with Uniform Cost Search (UCS) is the best choice because the robot does not know the complete environment before starting. It explores the building step by step, updates its knowledge whenever it finds new obstacles or risky zones, and always chooses the least-cost path. This helps the robot reach the survivor safely and efficiently.